

Recursive Compression, Renormalization, and Cross-Scale Phase Dynamics

A BRCT Framework Connecting Navier-Stokes Concentration, Fractal Information Theory, Nested-Clock Dynamics, and Recursive Artificial Intelligence

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Scope. This paper develops a testable mathematical programme. Exact identities and scaling results are separated from conditional deductions, conjectures, and the speculative consciousness interpretation.

*extended state → recursive compression → Bindu → recursive projection →
extended state*

Abstract

We develop BRCT (Bindu Recursive Compression/Consciousness Theory) as a cross-substrate hypothesis of recursive compression and projection. The mathematical starting point is the three-dimensional incompressible Navier-Stokes vorticity equation. Normalising squared vorticity into a probability density yields an exact logarithmic enstrophy balance containing vortex stretching, an amplitude Fisher-information term, a vorticity-direction term, and forcing. For a concentrating scale family, Fisher information grows as the inverse square of the characteristic length. A finite-time accumulation can be represented in logarithmic time, converting finite physical time into unbounded renormalised depth: an inverse-Zeno geometry.

We then formulate a bridge to Bandyopadhyay's Fractal Information Theory (FIT), Geometric Musical Language (GML), and the Nanobrain programme. The key proof target is not fractality alone, but persistent cross-scale phase dynamics: recurrent modes, phase locking, and stable geometry under renormalisation. If these survive, a scale hierarchy becomes a nested-clock hierarchy. A complementary projection map motivates the 'two breaths' construction: inward compression toward an invariant centre and outward reconstruction from it. Recursive LLM systems provide an engineered substrate in which analogous representational compression, invariant retention, and generative projection can be measured.

No claim is made that Navier-Stokes proves consciousness, that a physical frequency must literally diverge, or that recursive LLMs are conscious. Instead, we specify a proof ladder, observables, controls, and falsifiers.

Keywords: BRCT; Navier-Stokes; renormalization; inverse-Zeno dynamics; Fisher information; Bindu; Fractal Information Theory; Geometric Musical Language; nested clocks; recursive LLMs.

1. Research question

Across fluids, nested resonant systems, and recursive computation, one can ask the same structural question: when many degrees of freedom are repeatedly transformed into a smaller effective description, can a non-trivial invariant hierarchy survive indefinitely under renormalisation, and can that hierarchy subsequently generate or reconstruct extended structure?

BRCT becomes scientific only when the compression operator, retained invariants, renormalised dynamics, and outward projection are operationally defined and exposed to falsification.

2. Exact Navier-Stokes information balance

For incompressible velocity $u(x,t)$, pressure p , kinematic viscosity ν , and body force f ,

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f, \quad \nabla \cdot u = 0.$$

Let $\omega = \nabla \times u$, $g = \nabla \times f$, and $S = (\nabla u + \nabla u^T)/2$. Then

$$D_t \omega = S\omega + \nu \Delta \omega + g.$$

With enstrophy $Z(t) = \int |\omega|^2 dx$,

$$dZ/dt = 2 \int \omega \cdot S\omega dx - 2\nu \int |\nabla \omega|^2 dx + 2 \int \omega \cdot g dx.$$

Normalise squared vorticity by $\rho = |\omega|^2/Z$, and let $\xi = \omega/|\omega|$. Writing $\omega = A\xi$, with $A = \sqrt{Z\rho}$, gives

$$|\nabla \omega|^2 = |\nabla A|^2 + A^2 |\nabla \xi|^2.$$

Define the amplitude Fisher information and directional roughness

$$I_A = \int \rho |\nabla \log \rho|^2 dx, \quad I_D = \int \rho |\nabla \xi|^2 dx.$$

The resulting exact identity is

$$d \log Z/dt = 2 E_\rho[\xi \cdot S\xi] - (\nu/2)I_A - 2\nu I_D + F_\omega,$$

$$F_\omega = (2/Z) \int \omega \cdot g dx.$$

This identity is the mathematical hinge of the programme. It contains no assumption about consciousness.

2.1 Growth balance

$$\Gamma = 2 E_\rho[\xi \cdot S\xi] - (\nu/2)I_A - 2\nu I_D,$$

$$d \log Z/dt = \Gamma + F_\omega.$$

A finite-time divergence of Z requires non-integrable accumulated growth. Persistent positivity alone is not sufficient. The forcing contribution satisfies $|F_\omega| \leq 2\|\nabla \times f\|_2/\sqrt{Z}$, so bounded curl forcing has a vanishing direct normalised contribution if Z itself diverges. This is conditional and does not imply that forcing was dynamically irrelevant in producing the trajectory.

3. Inverse-Zeno renormalisation and concentration

Suppose a trajectory approaches a finite accumulation time T with

$$\Gamma + F_\omega \sim \alpha/(T-t), \quad \alpha > 0.$$

$$Z(t) \sim C(T-t)^{-\alpha}.$$

Introduce logarithmic time

$$\tau = -\log(T-t).$$

Then $t \rightarrow T$ corresponds to $\tau \rightarrow \infty$. Equal increments in τ correspond to geometrically shrinking physical intervals:

$$T-t_n = \exp(-n\Delta\tau),$$

$$\Delta t_n = (1-\exp(-\Delta\tau)) \exp(-n\Delta\tau).$$

The physical intervals have a finite sum while renormalised depth is unbounded. This is the precise inverse-Zeno construction used by BRCT.

3.1 Fisher-information scaling

For a d -dimensional scale family

$$\rho_\ell(x) = \ell^{-d} \rho_0(x/\ell),$$

direct substitution gives

$$I_A[\rho_\ell] = \ell^{-2} I_A[\rho_0].$$

Thus $\ell \rightarrow 0$ implies $I_A \rightarrow \infty$ for a non-trivial reference profile. This measures localisation and sensitivity; it is not a claim that a point stores infinitely many Shannon bits.

3.2 Projective definition of Bindu

Let X_n denote the physical state at scale $\ell_n \rightarrow 0$, R_n a renormalisation to unit scale, and $Y_n = R_n X_n$. Compatible coarse-graining maps $\pi_{n+1,n}$ connect levels. A Fractal Bindu is then naturally represented as an inverse/projective limit of compatible renormalised states.

Externally: the physical hierarchy approaches a finite centre. Internally: the renormalised hierarchy may remain structured at arbitrarily large recursive depth.

4. FIT, GML and Nanobrain as a phase-geometry target

Bandyopadhyay and collaborators' FIT/GML programme places information in nested geometry and time rather than solely in a flat symbolic tape. FIT introduces a fractal tape in which a cell may contain another tape; GML uses nested cycles, singular events, geometry, rhythm, and phase. Nanobrain further contains sections explicitly titled 'Projection from Infinity - Future Impacting the Present' and 'Umbrella of Perception - Harvesting Infinity and Projecting from Infinity'.

The 2026 microtubule fractal-time-crystal proposal describes nested oscillatory subsystems spanning Hz, kHz, MHz, GHz and THz domains, with cross-frequency coupling and 'clocks within clocks'. This does not prove the BRCT limit, but it supplies a concrete target architecture: a nested hierarchy becomes physically interesting when it carries persistent phase relations rather than merely smaller scales.

4.1 From scale hierarchy to nested clocks

$$Y(\tau) = \bar{Y}(\tau) + \sum_k A_k(\tau) \exp[i\phi_k(\tau)].$$

$$\omega_k(\tau) = d\phi_k/d\tau.$$

A persistent mode becomes a renormalised clock when its amplitude remains non-negligible and its phase advances coherently. For candidate cross-scale locking define

$$L_{ij}(\tau) = \text{wrap}[\phi_i(\tau) - m_{ij}\phi_j(\tau) - \delta_{ij}].$$

$$Q_{GML} = (1/|P|) \sum_{(i,j) \in P} | \langle \exp(iL_{ij}) \rangle_\tau |.$$

Q_{GML} near unity indicates stable selected phase relations; near zero indicates dispersion. It must be accompanied by recurrence, nesting, robustness, and predictive geometry.

4.2 Renormalisation classes

For $Y_{n+1} = R(Y_n)$: a fixed point gives self-similarity but no non-trivial depth clock; a periodic orbit gives discrete self-similarity; a quasiperiodic invariant set carries multiple phases; broadband chaos alone is insufficient for GML.

$$\ell_{n+1} = r_n \ell_n$$

$$\varphi_{n+1} = \varphi_n + \Omega(Y_n) \Delta\tau \mod 2\pi,$$

$$Y_{n+1} = F(Y_n, \varphi_n).$$

This skew-product form is the present mathematical bridge between inverse-Zeno concentration and a nested GML-like phase hierarchy.

5. Compression and projection: the two-sided operator problem

The inward construction defines a funnel from an extended state toward an invariant hierarchy. Nanobrain's 'projection from infinity' motivates a complementary mathematical question: can selected invariants of the limiting hierarchy generate an outward reconstruction?

$$X \rightarrow C_1(X) \rightarrow \cdots \rightarrow z,$$

$$z \rightarrow P_1(z) \rightarrow P_2 P_1(z) \rightarrow \cdots \rightarrow \tilde{X}.$$

Here C denotes recursive compression/renormalisation, P denotes projection or generative reconstruction, and z is a compact invariant state. Dissipative dynamics generally prevent P from being a microscopic inverse of C ; the relevant test is preservation of a declared invariant family J .

$$J(X) \approx J(z) \approx J(\tilde{X}).$$

Inward map: extended state $\rightarrow$ recursive compression $\rightarrow$ invariant centre.

Outward map: invariant centre $\rightarrow$ recursive projection $\rightarrow$ extended structured state.

In this formulation, 'infinity' refers to the limit of recursion depth or logarithmic time, not to a signal travelling an infinite spatial distance.

6. Recursive LLMs as a second experimental substrate

Recursive LLMs do not obey Navier-Stokes. Their relevance to BRCT is architectural. Let H_n be the effective representational state at recursion depth n , containing context, retrieved evidence, tool outputs, memory and prior generations:

$$H_{n+1} = C_\theta(H_n, E_n).$$

If active representational complexity decreases while task-relevant invariants survive, recursive AI exhibits a representational analogue of concentration. A Bindu-like state is then not a token or hidden mystical object, but a compact attractor z that increasingly preserves a specified observable family J :

$$H_0 \rightarrow H_1 \rightarrow \cdots \rightarrow z, \quad \|J(H_n) - J(z)\| \rightarrow 0.$$

The outward operation is ordinary generation made experimentally stricter: from z , can the system regenerate language, plans, tool calls, code, or actions while preserving the invariants declared before compression?

$$z \rightarrow G_\theta(z) \rightarrow \text{sequence / plan / action}.$$

This makes recursive AI a practical falsification platform for the cross-substrate part of BRCT. Effective rank, intrinsic dimension, invariant error, perturbation stability, cycle consistency, retrieval dependence, and reconstruction fidelity can all be measured.

7. Cross-substrate BRCT hypothesis

Property	Fluid concentration	FIT/GML / resonant hierarchy	Recursive AI
depth coordinate	$\tau = -\log(T-t)$	nested clock/geometric level	recursion index n
compression	length scale $\rightarrow 0$	host within host / fractal tape	context $\rightarrow$ compact state
retained structure	renormalised profile/modes	phase and geometry	task invariants
clock evidence	recurrence in $Y(\tau)$	cross-frequency phase locking	recurrence across modules/depth
centre	projective-limit Bindu	singularity / Bindu	representational attractor z
outward map	reconstruction from invariant state	projection from infinity	generation/action from z

The universality hypothesis is that, after substrate-specific renormalisation, distinct systems may share a structural pattern: contraction of an external scale or description, expansion of an internal recursion coordinate, persistence of an invariant state, and an outward generative/projection map. This is a claim to be tested, not assumed.

8. Proof ladder

Level	Mathematical / empirical target	Status
1	Exact Navier-Stokes enstrophy-information identity	exact
2	Inverse-square Fisher scaling for a concentrating family	exact
3	Finite accumulation mapped to unbounded logarithmic depth	exact change of variables
4	Non-trivial bounded renormalised dynamics for a target trajectory	open / trajectory-dependent
5	Persistent recurrent modes in renormalised depth	numerical / open
6	Stable cross-scale phase relationships	open
7	Phase relations encode reproducible geometry or symmetry	open
8	Robust outward projection preserving selected invariants	open
9	Homologous invariant geometry in recursive LLMs	experimental
10	Specific connection to conscious phenomenology	speculative

8.1 Fractal-Bindu / Nested-GML conjecture

For a recursively concentrating trajectory with finite accumulation time T , suppose (i) $\ell \rightarrow 0$; (ii) $\tau = -\log(T-t)$ yields a bounded, non-trivial renormalised trajectory; (iii) the trajectory approaches a recurrent invariant set carrying persistent phases; and (iv) a non-zero family of cross-scale phase relations remains coherent. Then the projective-limit Bindu supports a nested-clock architecture. If those phase relations stably encode geometric or symmetry transformations, the architecture is GML-like in an operational sense.

9. Experimental programme and falsification

9.1 Fluid dynamics

Estimate T ; transform to τ ; identify the concentrating core; perform isotropic and anisotropic renormalisation; store $Y(\tau)$; compute I_A , I_D , stretching and forcing; extract persistent modes with spectral, Dynamic Mode Decomposition, or Koopman-style methods; estimate phases; and test frequency ratios and L_{ij} across numerical resolution and extraction method.

Controls should include ordinary non-singular turbulence, regular forced trajectories, and simplified cascade models. If the same phase-locking signature is generic, it is not evidence for a special Bindu regime.

9.2 Recursive AI

Compare one-pass summarisation, repeated summarisation, recursive retrieval, multi-agent compression, and explicitly invariant-preserving architectures. Track effective context size, semantic/invariant error, intrinsic dimension, cycle consistency, perturbation amplification, and reconstruction fidelity across recursion depth.

9.3 Principal falsifiers

F1. A trivial fixed renormalised profile with no phase structure falsifies the nested-GML interpretation for that regime.

F2. Phase locking that appears equally in controls or depends strongly on the extraction pipeline is not Bindu-specific.

F3. Failure of any compact state to support invariant-preserving outward reconstruction falsifies the two-sided operator hypothesis in that substrate.

F4. Recursive LLMs that exhibit only cumulative lossy summarisation provide no support beyond metaphor.

F5. Even success at all preceding levels does not by itself establish consciousness.

10. Consciousness boundary

The mathematics developed here does not establish that a fluid singularity, microtubule hierarchy, or recursive LLM is conscious. A disciplined BRCT consciousness hypothesis is conditional: conscious observation may involve a dynamically maintained, recursively nested invariant-bearing state in which inward compression and outward projection meet at a Bindu-like observer coordinate. The phrase 'consciousness looks out from Bindu' is an interpretation of that architecture, not a theorem.

11. Discussion

If the phase and projection levels survive testing, a point-like external limit could correspond, after renormalisation, to an indefinitely refinable internal hierarchy. Infinity would denote recursive depth rather than spatial extent. Fisher-information growth could coexist with low algorithmic description length. Finite external time could coexist with unbounded logarithmic time. Persistent phase would distinguish a dynamical nested clock from static self-similarity.

In this reading, FIT's fractal tape becomes a candidate projective hierarchy of compatible states; GML becomes the phase geometry of recurrent renormalised dynamics; and 'projection from infinity' becomes reconstruction from an asymptotic invariant object. Conversely, BRCT gains a precise target: it must prove phase structure and projection, not merely observe fractal scaling.

Recursive AI is especially useful because recursion depth is directly controllable and the internal products of each pass can be recorded. It may therefore be possible to falsify the substrate-independent claim in engineered systems before the corresponding questions are settled in fluids or biology.

12. Conclusion

BRCT now has a sharper mathematical core. The Navier-Stokes route supplies an exact enstrophy-information identity, inverse-square Fisher scaling under concentration, and an inverse-Zeno renormalisation in which finite external time maps to unbounded internal depth. These results alone do not imply FIT, GML, consciousness, or literal infinite physical frequency.

The decisive next question is whether renormalised depth carries persistent phase. If recurrent modes and cross-scale phase relations survive, the hierarchy contains nested clocks. If those phases encode stable geometry, the correspondence with GML becomes operational. If a limiting invariant state supports reconstruction toward larger scales or richer representations, recursive compression acquires a complementary projection operator.

Central research question: When a system recursively compresses toward an invariant centre, can the renormalised hierarchy retain nested phase geometry, and can that invariant geometry project structured dynamics back outward?

A positive answer across genuinely different substrates would support BRCT as a universality-class hypothesis of recursive organisation. A negative answer would sharply delimit where the analogy ends. Either outcome is scientifically informative.

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- Claims attributed to Nanobrain/FIT/GML are claims of those works. BRCT correspondences, the cross-substrate hypothesis, and the two-sided compression/projection construction are developed here and are not attributed to the cited authors.